Lecture 1: AWS Introduction

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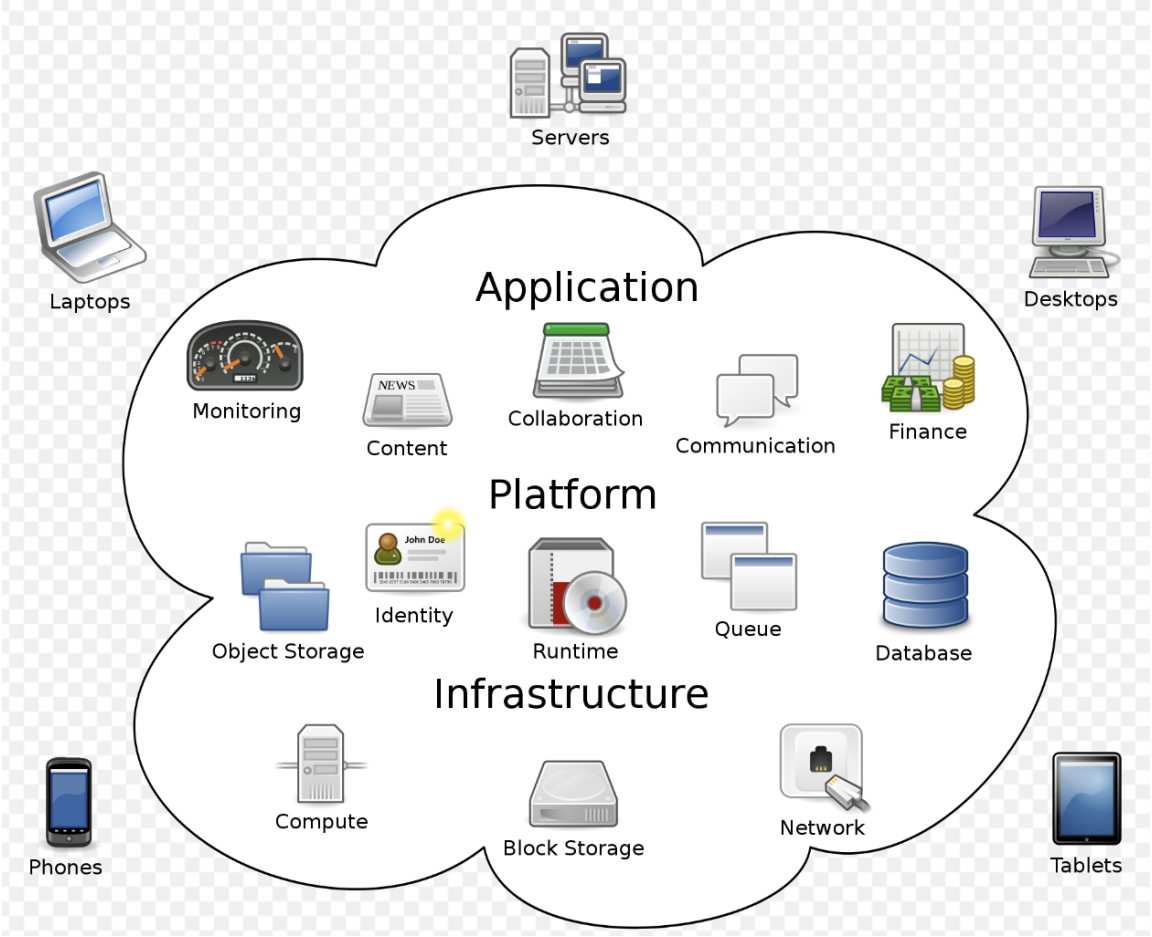
Agenda

- Cloud Computing
- AWS Overview

Cloud Computing

- The group of networked elements providing services does not need to be addressed and managed by users.

https://en.wikipedia.org/wiki/Cloud_computing



Cloud Computing

- Security
- Scalability and flexibility
- Cost Efficiency
- Ease of Access
- Disaster Recovery and Data Loss Prevention
- Automatic Software updates

Cloud Computing

- Major players
  - **AWS**
  - Microsoft Azure
  - Google Cloud Platform (GCP)
  - IBM Cloud
  - Oracle Cloud
  - Salesforce

Cloud computing SERVICE

- **Infrastructure** as a service
- **Platform** as a service
- **Software** as a service
- **Function** as a service
- **Container** as a service
- **Data** as a service
- **Desktop** as a service
- **Integration** as a service
- **Network** as a service
- **Security** as a service

Cloud Computing model



Hosted applications/apps	Development tools, database management, business analytics	Operating systems	Servers and storage	Networking firewalls/security	Data center physical plant/building

https://azure.microsoft.com/en-us/resources/cloud-computing-dictionary/what-is-paas/

Infrastructure as a Service (IaaS)

- Rent hardware such as servers from the cloud provider.
- Chose the operating system, memory, hard drive, and CPU size.
- Receive a credential to log in to your server after it is created . Once the server is provisioned, you can do whatever you want in the server such as hosting a website you developed.
- The IaaS service in the AWS cloud is an EC2.

Container as a Service

- Containerized deployments took over deployments on virtual machines. Because it is much lighter and faster to deploy apps.
- A Container as a Service (CaaS) model allows you to run containerized applications in the cloud.
- Containerized applications are platform-agnostic (cross-platform)
- Docker is the most popular containerization technology. In AWS, there 2 ways to run containerized applications, on servers (EC2 or EKS) or serverless (ECS Fargate).

Function as a Service (FaaS)

- FaaS allows customers to develop, run, and manage application functionalities **without the complexity** of building and maintaining the infrastructure and servers.
- Achieve a **serverless** architecture and is typically used when building modern event-driven and microservices applications.
- Serverless computing is a cloud computing execution model in which the cloud provider allocates machine resources on demand, taking care of the servers on behalf of their customers.
- The FaaS service in the AWS cloud is a Lambda.

	IaaS	CaaS	FaaS
The application runs on	Virtual Machines (VM)	Cluster of VMs managed by you or the cloud provider	In the cloud
Your responsibility	A lot of things, such as patching, networking, OS, runtime, and so on.	Container image	Application code
Scaling duration	Minutes	In a minute	In a second
Cost	A lot	Similar to IaaS	Least expensive when the number of calls is low.

Platform as a Service (PaaS)

- You don't know the cloud. And you've just got your code and want to run it in the cloud. Then use PaaS services. It will **provision** the underlying resources for you to run your code. You still have to look after the underlying assets, but you don't have to worry about provisioning of them. Great for traditional and monolith apps.
- PaaS is where you can directly drag and drop your (backend) app and hit deploy. Then your app is available publicly.
- The PaaS service in the AWS cloud is an Elastic Beanstalk. Under the hood, it utilizes the IaaS services such as EC2, Load Balancers, RDS.
- Other PaaS providers are Heroku, Vercel, Digital Ocean and so on.

Software as a Service (SaaS)

- SaaS is the applications and customers use it via the internet. You are not worried about the underlying servers, how their loads are balanced, high availability, DNS resolving, etc.
- For example, Amazon Rekognition is a cloud-based image and video analysis service that makes adding advanced computer vision capabilities to applications easy. Developers who do not have advanced machine learning knowledge can have AI features in the application.
- SaaS could be your application in the cloud running in the IaaS, PaaS, and FaaS models.

Cloud Computing Deployments

- Public Cloud
- Private Cloud
- Hybrid Cloud

AWS

- Amazon Web Service.
- The largest Cloud Computing provider.
- Provide a variety of scalable computing resources and services that can be used to build, deploy, manage applications on a global network.

Compute
- EC2
- Lightsail
- Lambda
- Batch
- Elastic Beanstalk
- Serverless Application Repository
- AWS Outposts
- EC2 Image Builder
- AWS App Runner

Containers
- Elastic Container Registry
- Elastic Container Service
- Elastic Kubernetes Service
- Red Hat OpenShift Service on AWS

Storage
- S3
- EFS
- FSx
- S3 Glacier
- Storage Gateway
- AWS Backup

Customer Enablement
- AWS IQ
- Support
- Managed Services
- Activate for Startups

Robotics
- AWS RoboMaker

Blockchain
- Amazon Managed Blockchain

Satellite
- Ground Station

Quantum Technologies
- Amazon Braket

Management & Governance
- AWS Organizations
- CloudWatch
- AWS Auto Scaling
- CloudFormation
- CloudTrail

Machine Learning
- Amazon SageMaker
- Amazon Augmented AI
- Amazon CodeGuru
- Amazon DevOps Guru
- Amazon Comprehend
- Amazon Forecast
- Amazon Fraud Detector
- Amazon Kendra
- Amazon Lex
- Amazon Personalize
- Amazon Polly
- Amazon Rekognition
- Amazon Textract
- Amazon Transcribe
- Amazon Translate
- AWS DeepComposer
- AWS DeepLens
- AWS DeepRacer
- AWS Panorama
- Amazon Monitron
- Amazon HealthLake
- Amazon Lookout for Vision
- Amazon Lookout for Equipment

AWS Cost Management
- AWS Cost Explorer
- AWS Budgets
- AWS Marketplace Subscriptions
- AWS Application Cost Profiler

Front-end Web & Mobile
- AWS Amplify
- Mobile Hub
- AWS AppSync
- Device Farm
- Amazon Location Service

AR & VR
- Amazon Sumerian

Application Integration
- Step Functions
- Amazon AppFlow
- Amazon EventBridge
- Amazon MQ
- Simple Notification Service
- Simple Queue Service
- SWF

AWS Infrastructure

- Compute: EC2, ECS, LightSail, Lambda, Batch
- Storage: EBS, EFS, S3, Glacier, Storage Gateway, Storage Migration Services
- Database
- Network (CloudFront, VPC, Direct Connect, Load Balancing, Route 53)

AWS Database

- Relational Database Service (RDS)
- Aurora: Relational DB Cluster
- DynamoDB: Nonrelational DB
- DocumentDB: Mongodb-compatible database cluster
- Neptune: Graph DB
- ElasticCache: Managed Redis and Memecached (in-memory data)

Security, Identity & compliance

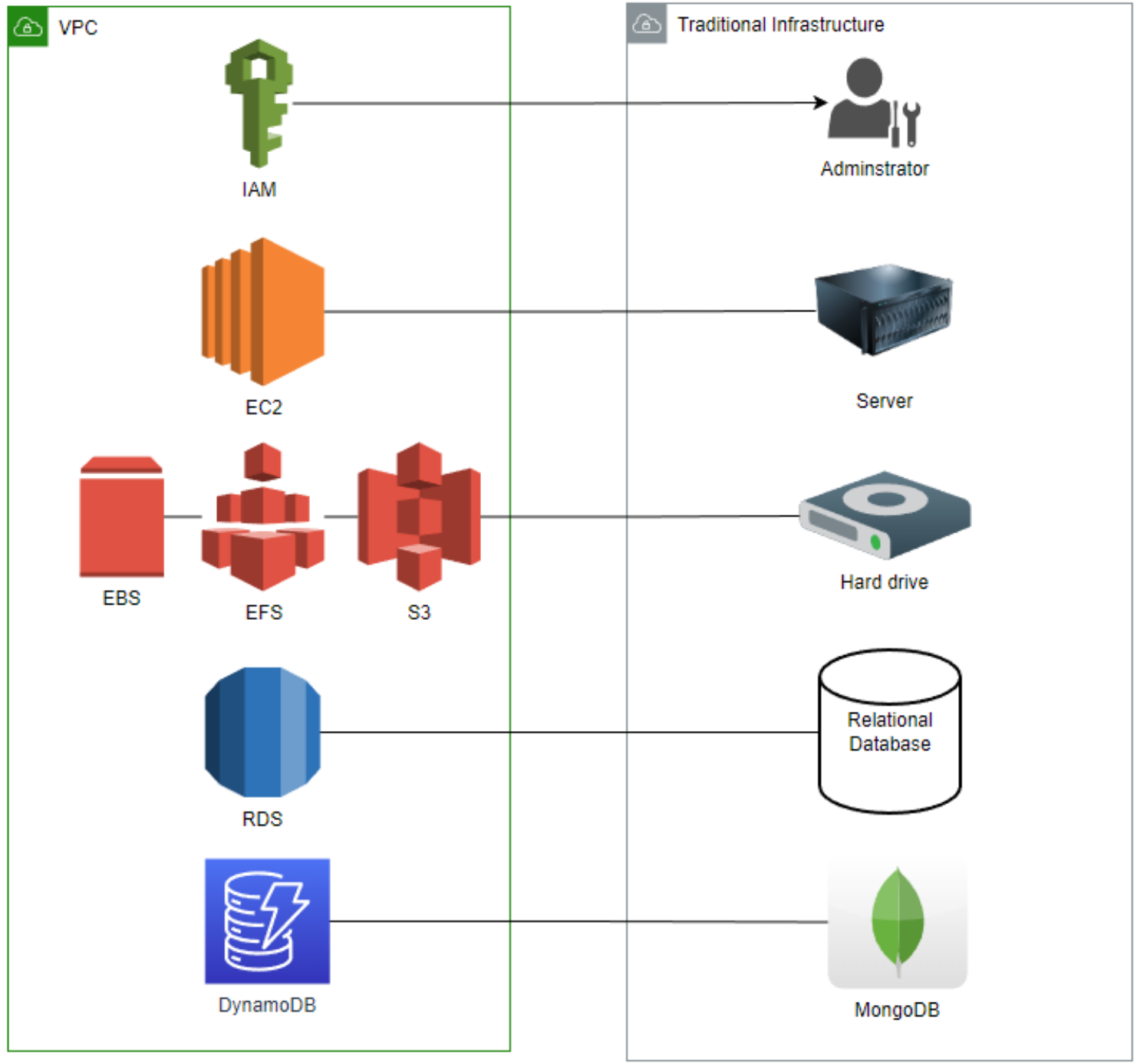
- IAM (identity and Access Management): manage users
- Cognition: authentication/OAuth service
- Certificate manager: Manage domain certificates registered via AWS/Route 53
- Guard Duty: Monitor malicious activity
- Inspector: an agent to test vulnerabilities
- Macie: Check the personal identifiable information (PII) in the entire application using ML

Security, Identity & compliance

- Cloud HSMHardware Security Module: a hardware containing private & public keys to access application/EC2 instances
- Directory Services: Similar to Microsoft Active Directory
- WAF (Web Application Firewall): Sit in front of web servers to protect websites from injection, cross-scripting.
- Shield: Prevent DDOS
- Artifact: For Compliance and Audit

Application Integration

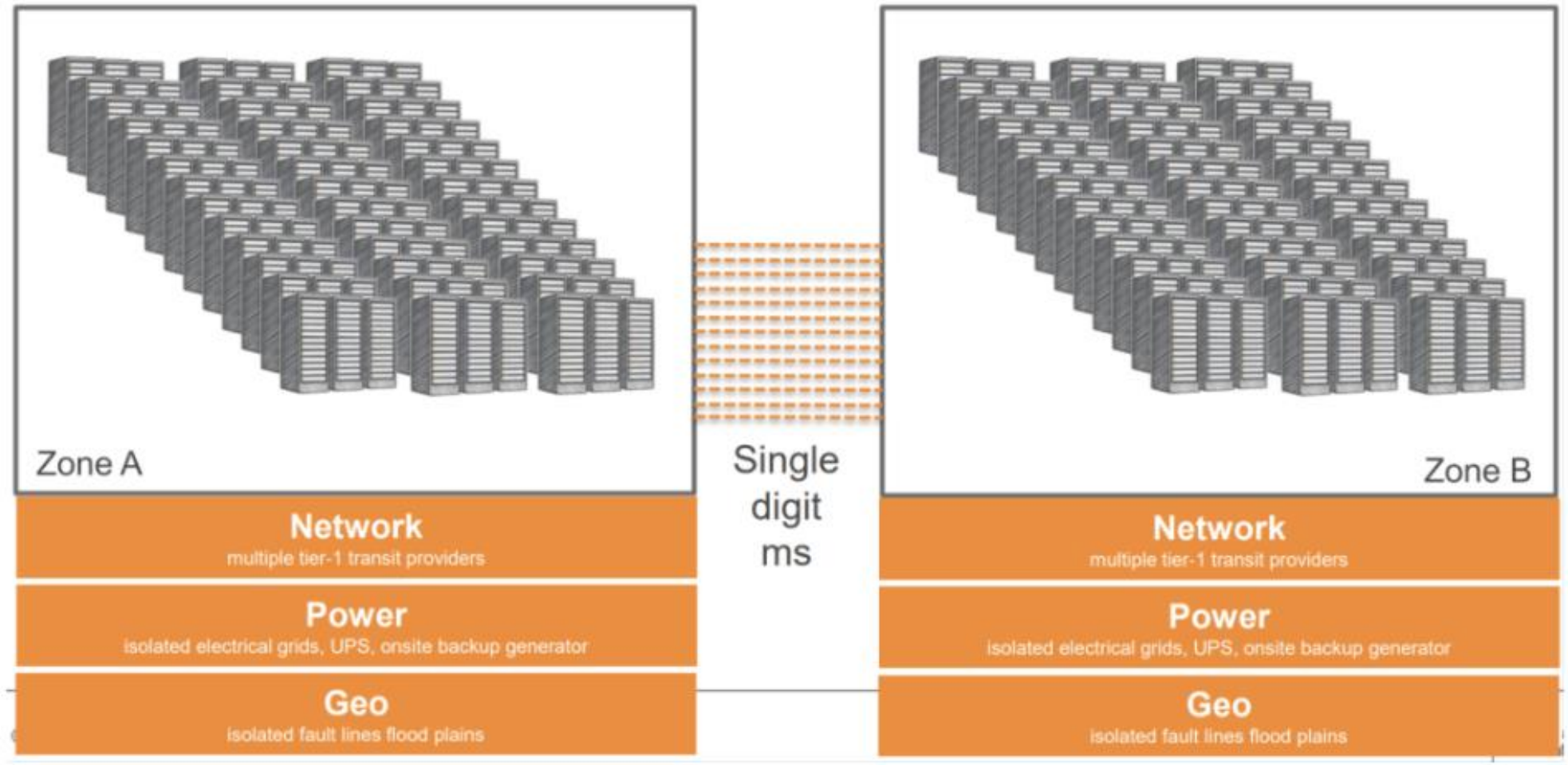
- Amazon MQ: Message broker (RabbitMQ)
- SNS (Simple Notification Service) (Push-based)
- SQS (Simple Queue Service) (Pull-based)
- SWF (Simple WorkFlow Service)


VPC
IAM
EC2
EBS
EFS
S3
RDS
DynamoDB
Traditional Infrastructure
Adminstrator
Server
Hard drive
Relational Database
MongoDB

Available Zones (AZ) & regions

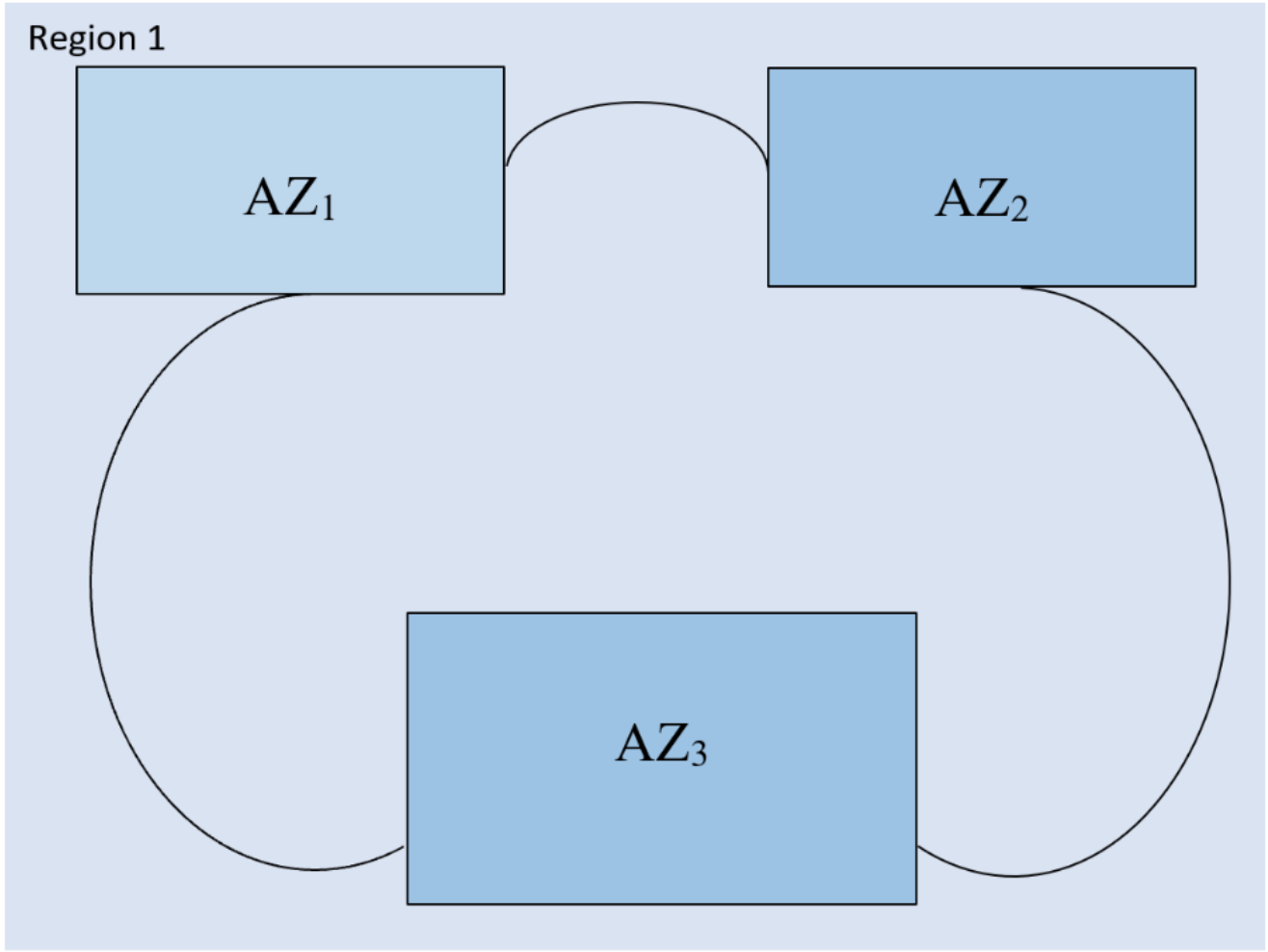
- A region is a geographically distinct area
- AZs are datacenters within a region.
- AWS has 16 regions and 44 AZs
- Each AZ is independent of other AZs

AZs and regions



Availability Zones are independent of other

Redundant AZs



Redundant AZs

Regular AWS accounts

- Homepage: https://aws.amazon.com/
- Full access to all services without limitation.
- Long-term access
- Production-level project
- Practical experience: Gain industry-like exposure.
- Credit/Debit card required.
- Free tier available: Up to 12 months for new accounts.
- Caution: If your usage exceeds the free tier limits, you may incur unexpected charges reflected in your credit card. You are solely responsible for this charge.

AWS Account

- Best practices to ensure cost-efficiency
  - Frequently monitor your resource usage.
  - Setup billing alert.
  - Check the bill daily.
  - Shutdown all unused instances/resources:
    - EC2 (Elastic Compute Cloud).
    - RDS (Relational Database Service), Aurora
    - ECS/EKS (Elastic Container Service/Elastic Kubernetes Service).
    - EBS (Elastic Block Store).
    - EFS (Elastic File System).
    - Unused Elastic IP addresses (EIP)
    - DocumentDB
    - DynamoDB.
    - CloudFront.
    - AWS Lambda.
    - SNS/SQS
    - VPC: NAT Gateway, Data transfer, VPN connections. You can keep the default VPC, which is already connected to the internet, so there is no need to create a new one when launching EC2 or other instances. Additionally, the default VPC does not incur significant cost.
    - S3: Particularly important when storing vast amounts of data.

Demo: S3

References

- https://docs.aws.amazon.com/
- ChatGPT: https://chatgpt.com/
- Google AI: https://gemini.google.com/app